

Q01.

```
Demo.java x
1  class Demo{
2      public static void printMyMethod(){
3          System.out.println("Hello JAVA");
4      }
5
6      public static void main(String[] args){
7          printMyMethod();
8      }
9  }
10
```

Output>>

```
Hello JAVA

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q02.

```
import java.util.*;
class Q2 {
    public static void personalInfor(){
        System.out.println("Tharush Wijemanna");
        System.out.println("2006/08/08");
    }

    public static void main(){
        personalInfor();
    }
}
```

Output>>

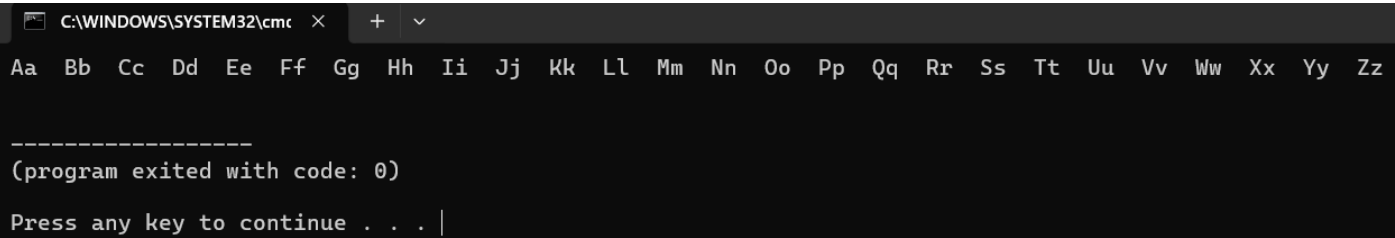
```
Tharush Wijemanna
2006/08/08

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q03.

```
class Q3 {  
    public static void printAlphabet(){  
        for(char upperchar = 'A', lowerchar = 'a' ; upperchar <= 'Z' ; upperchar++, lowerchar++){  
            System.out.print(upperchar+" "+lowerchar+" ");  
        }  
        System.out.println();  
    }  
    public static void main(String[] args){  
        printAlphabet();  
    }  
}
```

Output>>



```
C:\WINDOWS\SYSTEM32\cmd.exe  
Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

Q04.

```
import java.util.*;
class Q4 {
    public static void PrintString(){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter String : ");
        String text = input.nextLine();
        System.out.println(text);
    }
    public static void main(String[] args){

        PrintString();
    }
}
```

Output>>

```
Enter String : Rush
Rush

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q05.

```
import java.util.*;
class Q5 {
    public static void isleapyear(){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Year : ");
        int year = input.nextInt();
        if(year%4==0&&year%100!=0||year%400==0){

            System.out.println(year+" is leap Year");
        }else
            System.out.println(year+" is not Leap Year ");
    }
    public static void main(){
        isleapyear();
    }
}
```

Output>>

```
i
i Enter Year : 2008
2008 is leap Year
}
-----
(program exited with code: 0)
t
```

```
Enter Year : 2007
2007 is not Leap Year

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q06.

```
import java.util.*;
class Q6 {
    public static void integerCompare(){
        Scanner input = new Scanner(System.in);

        System.out.print("Input Number 1 - ");
        int num1 =input.nextInt();

        System.out.print("Input Number 2 - ");
        int num2 =input.nextInt();

        if(num1==num2){
            System.out.println("num1==num2 : "+num1+" = "+num2);
        }else if(num1>num2){
            System.out.println("--> "+num1+" > "+num2);
        }else{
            System.out.println("--> "+num1+" < "+num2);
        }
    }
    public static void main(String[] args){
        integerCompare();
    }
}
```

Output>>

```
Input Number 1 - 20
Input Number 2 - 25
--> 20 < 25
-----
```

```
Input Number 1 - 25
Input Number 2 - 20
--> 25 > 20
-----
Program exited with code 0
```

If the user input the same numbers . and output is given below

```
Input Number 1 - 20
Input Number 2 - 20
num1==num2 : 20 = 20
```

Q07.

07. Program 01

```
import java.util.*;
class Demo{
    public static void printName(){
        Scanner input=new Scanner(System.in);
        System.out.print("Input first name : ");
        String name1=input.nextLine();
        System.out.print("Input second name : ");
        String name2=input.nextLine();
        System.out.println(name1+" "+name2);
    }

    public static void main(String[] args){
        printName(); // Line X
    }
}
```

Program 02

```
import java.util.*;
class Demo{
    public static void printName(String name1, String name2){
        System.out.println(name1+" "+name2);
    }

    public static void main(String[] args){
        Scanner input=new Scanner(System.in);
        System.out.print("Input first name : ");
        String name1=input.nextLine();
        System.out.print("Input second name : ");
        String name2=input.nextLine();

        printName(name1,name2); // Line Y
    }
}
```

i.

*Program 01

It does not get arguments from the main method instead, it uses the scanner class inside the sub-method(printName()) to get user input and print it directly.

* program 02

It receives arguments passed from the main method(name1 and name2) and print them using those parameter values.

And these are the main differences between program 01 and program 02

ii.

Answer is : NO

In program 02 the printName method is defined as public static void printName(String name1, String name2) which requires two String parameters calling printName() .without passing

Any argument from main method does not match the method signature defined in program 02

Q08.

```
import java.util.*;
class Q8 {
    public static void checkFirstLetter() {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first word: ");
        String word1 = input.next();

        System.out.print("Enter second word: ");
        String word2 = input.next();

        char first1 = word1.charAt(0);
        char first2 = word2.charAt(0);

        if (first1 == first2 || first1 == first2 + 32 || first1 == first2 - 32) {
            System.out.println("true");
        } else {
            System.out.println("false");
        }
    }

    public static void main(String[] args) {
        checkFirstLetter();
    }
}
```

Output>>

```
Enter 1 Input - rush
Enter 2 Input - Rush
true

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
Enter 1 Input - Rush
Enter 2 Input - samith
false
```

Q09.

```
emo.java  Q2.java  Q3.java  Q4.java  Q5.java  Q6.jav
import java.util.*;
class Q9 {
    public static void printRound(int num){
        System.out.printf("%.2f \n ",(double)num);
    }
    public static void main(String[] args){
        Scanner input =new Scanner(System.in);

        System.out.print("Enter Number : ");
        int num = input.nextInt();
        printRound(num);
    }
}
```

Output>>

```
Enter Number : 23.45544
23.46

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q10.

```
no.java  Q2.java  Q3.java  Q4.java  Q5.java  Q6.java  Q8.java  Q9.java  Q10.java
import java.util.*;
class Q10 {
    public static void getrequired(){
        Scanner input =new Scanner(System.in);

        System.out.print("Enter desired monthly interest - Rs.");
        double m_interest = input.nextDouble();

        double annual_interest =m_interest*12;
        double annualrate = 0.2;
        double required_deposite = annual_interest/annualrate;
        System.out.printf("Required deposit amount : Rs.%.2f ",required_deposite);
    }
    public static void main(String[] args){
        getrequired();
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter desired monthly interest - Rs.20000
Required deposit amount : Rs.1200000.00

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q11.

```
import java.util.*;
class Q11 {
    public static void getconvert(double c_val){

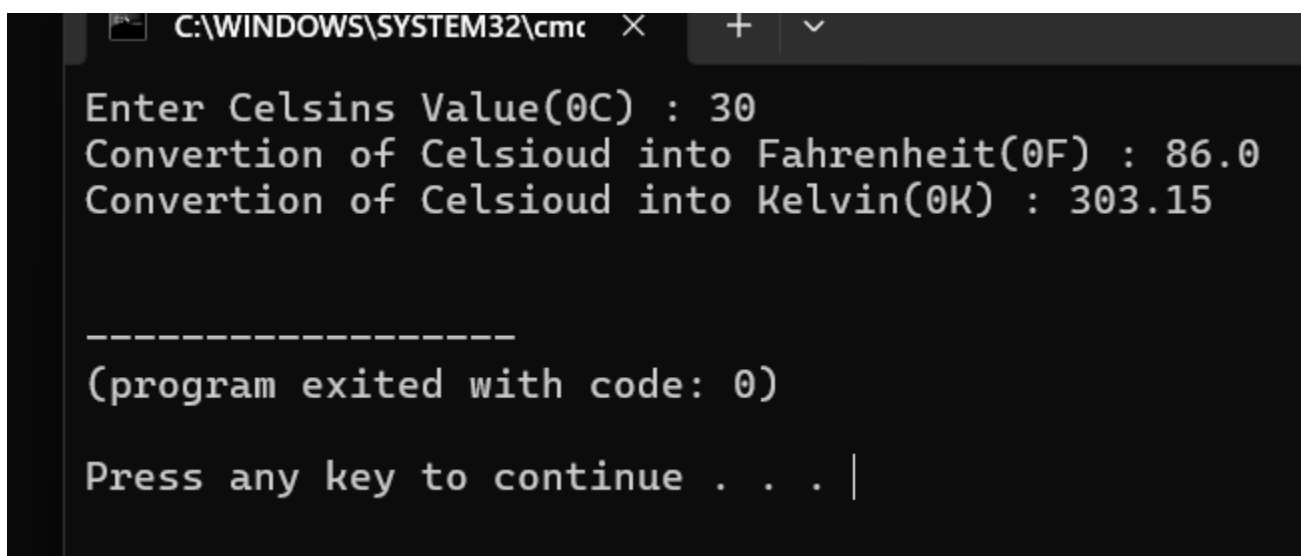
        double f_val = (c_val*9/5)+32;
        double k_val = c_val+273.15;

        System.out.println("Conversion of Celsioud into Fahrenheit(0F) : "+f_val);
        System.out.println("Conversion of Celsioud into Kelvin(0K) : "+k_val);
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter Celsins Value(0C) : ");
        double c_val = input.nextDouble();

        getconvert(c_val);
    }
}
```

Output>>



```
C:\WINDOWS\SYSTEM32\cmd
Enter Celsins Value(0C) : 30
Conversion of Celsioud into Fahrenheit(0F) : 86.0
Conversion of Celsioud into Kelvin(0K) : 303.15

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q12.

```
import java.util.*;
class Q12 {
    public static void getcalculate(){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter angle (in degrees) : ");
        double angle_value =input.nextDouble();

        System.out.print("Enter radius (cm) : ");
        double radius =input.nextDouble();
        double arc_length =(angle_value/360)*(2*(3.14*radius));
        double perimeter = arc_length+(2*radius);
        double area_sector = (angle_value/360)*(3.14*radius*radius);

        System.out.printf("Perimeter : %.2f \n",perimeter);
        System.out.printf("Area of a sector : %.2f \n",area_sector);
    }
    public static void main(String[] args){

        getcalculate();

    }
}
```

Output>>

```
Enter angle (in degrees) : 90
Enter radius (cm) : 14
Perimeter : 49.98
Area of a sector : 153.86
```

(program exited with code: 0)

Press any key to continue . . . |

Q13.

```
import java.util.*;
class Q13 {
    public static void findquadratic(){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter first integer for x(a) : ");
        int num1 = input.nextInt();

        System.out.print("Enter second integer for x(b) : ");
        int num2 = input.nextInt();
        int sum = num1+num2;
        int multipli = num1*num1;
        System.out.println("x^2 + "+sum+"x "+multipli+ " = 0");

    }
    public static void main(String[] args){
        findquadratic();
    }
}
```

Output>>

```
Enter first integer for x(a) : 4
Enter second integer for x(b) : 7
x^2 + 11x +16 = 0
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

Q14.

```
import java.util.*;
class calculator {
    public static void add(int a,int b){
        System.out.println(a + " + " +b+" = "+(a+b));
    }
    public static void subtraction(int a,int b){
        System.out.println(a + " - " +b+" = "+(a-b));
    }
    public static void multiplication(int a,int b){
        System.out.println(a + " * " +b+" = "+(a*b));
    }
    public static void dividend(int a,int b){
        double divi = (double) a/b;
        if(a>b){
            System.out.printf(a + " / " +b+" = %.2f \n ",divi);
        }else{
            System.out.printf(a + " / " +b+" = %.2f \n ",divi);
        }
    }
    public static void reminder(int a,int b){
        int remin = a%b;
        System.out.println(a + " % " +b+" = " +remin);
    }
    public static void power(int a,int b){
        double power_0 = Math.pow(a,b);
        System.out.printf(a+" ^ "+b+" = %.0f",power_0);
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);
        System.out.print("Enter the first number : ");
        int num1 =input.nextInt();

        System.out.print("Enter the second number : ");
        int num2 =input.nextInt();

        System.out.print("Enter the operator (+,-,*,/,%,^) : ");
        char operator = input.next().charAt(0);

        switch(operator) {
            case '+':
                add(num1,num2);
                break;
            case '-':
                subtraction(num1,num2);
                break;
            case '*':
                multiplication(num1,num2);
                break;
            case '/':
                dividend(num1,num2);
                break;
            case '%':
                reminder(num1,num2);
                break;
            case '^':
                power(num1,num2);
                break;
        }
    }
}
```

Output>>

```
01 pr
02 Enter the first number : 20
03 Enter the second number : 32
04 Enter the operator (+,-,*,/,%,^) : +
05 20 + 32 = 52
06 -----
07
```

```
08 fiv
09 fou
10 Enter the first number : 20
11 Enter the second number : 56
12 Enter the operator (+,-,*,/,%,^) : -
13 20 - 56 = -36
14 -----
15
```

```
16
17 Enter the first number : 34
18 Enter the second number : 5
19 Enter the operator (+,-,*,/,%,^) : *
20 34 * 5 = 170
21 -----
22
```

```
23 C:\WINDOWS\system32\cmd.exe
24 mo
25 Enter the first number : 45
26 Enter the second number : 3
27 Enter the operator (+,-,*,/,%,^) : /
28 45 / 3 = 15.00
29 -----
30
```

```
31 no
32 s
33 c
34 Enter the first number : 45
35 Enter the second number : 4
36 Enter the operator (+,-,*,/,%,^) : %
37 45 % 4 = 1
38 -----
39
```

```
40 Enter the first number : 5
41 Enter the second number : 2
42 Enter the operator (+,-,*,/,%,^) : ^
43 5 ^ 2 = 25
44 -----
45 (program exited with code: 0)
```

Q15.

```
class Q15 {  
    public static void printswap(){  
        int a =23;  
        int b =26;  
        a= a+b;  
        b= a-b;  
        a= a-b;  
        System.out.println("After Swaping : \n a = "+a+ "\n b = "+b);  
    }  
    public static void main(String[] args){  
        printswap();  
    }  
}
```

Output>>

```
After Swaping :  
a = 26  
b = 23  
-----
```

Q16.

```
Q3.java ✕ Q4.java ✕ Q5.java ✕ Q6.java ✕ Q8.java ✕ Q9.java ✕ Q10.java
import java.util.*;
class Q16 {
    public static void printperfect(){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter Number : ");
        int num = input.nextInt();

        if(num<=0){
            System.out.println("Error");
        }else{

            int sum = 0;

            for(int i = 1;i<num;i++){
                if(num%i==0){
                    sum+=i;
                }
            }

            if(sum==num){
                System.out.println(num+" is a Perfect Number.");
            }else{
                System.out.println(num+" is not a Perfect Number.");
            }
        }
    }

    public static void main(String[] args){
        printperfect();
    }
}
```

Output>>

```
Enter Number : 28
28 is a Perfect Number.

-----
(program exited with code: 0)
```

Q17.

```
Q5.java Q6.java Q8.java Q9.java Q10.java Q11.java Q12.java Q13.java calculator.java Q15.java
import java.util.*;
class Q17 {
    public static void checkRange(){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter number : ");
        int num = input.nextInt();
        System.out.print("The lower bound of the range : ");
        int lower_bound=input.nextInt();
        System.out.print("The upper bound of the range : ");
        int upper_bound=input.nextInt();
        int count =0;
        for(int i=lower_bound+1;i<upper_bound;i++){
            if(i%num==0){
                count++;
            }
        }
        System.out.println("Multiples of "+num+" between "+lower_bound+" and "+upper_bound+" - "+count);
    }
    public static void main(String[] args){
        checkRange();
    }
}
```

Output>>

```
Enter number : 3
The lower bound of the range : 10
The upper bound of the range : 100
Multiples of 3 between 10 and 100 - 30

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q18.

```
class Demo {  
    public static void test(){  
        System.out.println("Hello...");  
    }  
    public static void test(String name){  
        System.out.println("Hello "+name+"...");  
    }  
    public static void main(String args[]){  
        String s="java"; //Line 1  
        //test; //Line 2  
        test(); //Line 3  
        //test(){ } //Line 4  
        //test(){ }; //Line 5  
        Demo.test(); //Line 6  
        System.out.println("test()"); //Line 7  
        //System.out.println(test()); //Line 8  
        test(s); //Line 9  
        //System.out.println(test(s)); //Line 10  
        System.out.println("test(s)"); //Line 11  
    }  
}
```

Answers:

line 1 – legal

line 2 – illegal

* curly braces are required when calling a method

line 3 – legal

line 4 – illegal

* We can't declare a method inside the main method block.

line 5 – illegal

* We can't declare a method inside the main method block.

line 6 – legal

line 7 – legal

line 8 – illegal

* The test() can't be used inside System.out.println().

line 9 – legal

line 10 – illegal

* * The test(string name) can't be used inside System.out.println().

line 11 – legal

Q19.

```
import java.util.*;
class Q19 {
    public static void selectRandom(){
        Scanner input =new Scanner(System.in);

        System.out.print("Enter name : ");
        String name =input.nextLine();

        String[] group01 = {
            "Business & Accounting","Geography","Citizenship Education ","Entrepreneurship studies","2nd Language Sinhala",
            "2nd Language Tamil","Foreign Languages"
        };

        String[] foreignLanguage = {"Arabic","Hindi","French","Japan"};

        String[] group02 = {"Art","Tamil Literature","English Literature","Sinhala Literature","Music","Dancing" };

        String[] group03 = {"Information & technology","Agriculture","Home Economics","Health Science","Art & Craft","Media"};

        Random r =new Random();

        String subject1 = group01[r.nextInt (group01.length)];
        if(subject1==group01[6]){
            String selectlanguage = foreignLanguage[r.nextInt(foreignLanguage.length)];
            subject1= "Foreign Language : "+selectlanguage;
        }
        String subject2 = group02[r.nextInt (group02.length)];
        String subject3 = group03[r.nextInt (group03.length)];

        System.out.println(name+" has selected random subject in the system \n ");
        System.out.println("                First Subject: "+subject1);
        System.out.println("                Second Subject: "+subject2);
        System.out.println("                Third Subject: "+subject3);

    }
    public static void main(){
        selectRandom();
    }
}
```

Output>>

```
Enter name : some
some has selected random subject in the system

                First Subject: Foreign Language : French
                Second Subject: Sinhala Literature
                Third Subject: Health Science
```

(program exited with code: 0)

Press any key to continue . . . |

Q20.

```
class Test{
    public static void testMethod(){
        System.out.println("testMethod of Test");
    }
}

class Demo{
    public static void testMethod(){
        System.out.println("testMethod of Demo");
    }
    public void testMethod2(){
        System.out.println("testMethod of Demo");
    }
    public static void main(String args[]){
        testMethod(); //line 1
        //testMethod2(); //line 2
        Test.testMethod(); //line 3
        Demo.testMethod(); //line 4
        //Demo.testMethod2(); //line 5
        Demo d=new Demo(); //line 6
        d.testMethod(); //line 7
        d.testMethod2(); //line 8
    }
}
```

Answers:

Line 2 and Line 5 Shows compile error because testMethod2()/Demo.testMethod2 are non static, and static method can't be directly call non static methods.

Q21.

Answers

a. Legal

b. Legal

c. Illegal

* method body { } is missing

d. Illegal

* missing method identifier(name)

e. Illegal

* method body { } is missing

f. legal

g. Illegal

* we can't use semicolons when passing parameters

h. Illegal

* Missing data type for parameter(x)

i. illegal

* Missing data type and variable name

j. illegal

k. illegal

* without condition it was illegal

l. legal

Q22.

```
import java.util.*;
class Q22 {
    public static double convertSpeed(double x){

        double conver_meter = x*1000;
        int minute_insecond = 3600;

        double convert_speed = conver_meter/3600;

        return convert_speed;
    }
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter Speed (Kmh-1) : ");
        double speed =input.nextDouble();
        double c_speed=convertSpeed(speed);
        System.out.printf(speed+" Kmh-1 Convert to ms-1 : %.2f ms-1 ",c_speed);

    }
}
```

Output>>

```
Enter Speed (Kmh-1) : 160
160.0 Kmh-1 Convert to ms-1 : 44.44 ms-1
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q23.

```
import java.util.*;
class Q23 {
    public static int vowelCount(String word){

        int vowel_count = 0;
        for(int i=0;i<word.length();i++){
            char ch = word.charAt(i);
            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
               ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U'){

                vowel_count++;
            }
        }
        return vowel_count;
    }
    public static void main(){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter String Word : ");
        String in_word = input.nextLine();
        int v_count = vowelCount(in_word);
        System.out.println("There are "+v_count+" vowels in this word");
    }
}
```

Output>>

```
Enter String Word : Rush Wijemanna
There are 5 vowels in this word
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

Q24.

```
import java.util.*;
class Q24 {
    public static double tankCapacity(double radius, double height){
        double conv_Radius = radius*100;
        double conv_Height = height*100;

        double capacity_valu = (3.14*conv_Radius*conv_Radius)*conv_Height/1000;

        return capacity_valu;
    }
    public static void main(){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter Radius in Meter (m) : ");
        double r_valu = input.nextDouble();
        System.out.print("Enter Heigth in Meter (m) : ");
        double h_valu = input.nextDouble();

        double c_valu=tankCapacity(r_valu,h_valu);
        System.out.printf("Tank Capactiy : %.2f L",c_valu);
    }
}
```

Output>>

```
Enter Radius in Meter (m) : 3
Enter Heigth in Meter (m) : 5
Tank Capactiy : 141300.00 L

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q25.

```
Q25.java ~ Q9.java ~ Q10.java ~ Q11.java ~ Q12.java ~ Q13.java ~ Calculator.java ~ Q15.java
import java.util.*;
class Q25 {
    public static void getvalues(){
        Scanner input =new Scanner(System.in);

        System.out.print("Enter cube length (cm) : ");
        double length =input.nextDouble();

        double radius = length/2;
        double volume = 4/3*(3.14*radius*radius);

        System.out.printf("Volume of the largest sphere : %.2f cubic units ",volume);
    }
    public static void main(){
        getvalues();
    }
}
```

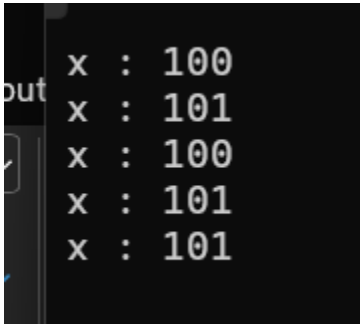
Output>>

```
C:\WINDOWS\SYSTEM32\cmd  ×  +  v
Enter cube length (cm) : 24
Volume of the largest sphere : 452.16 cubic units

-----
6 (program exited with code: 0)

Press any key to continue . . . |
```

Q26.



```
out x : 100
x : 101
x : 100
x : 101
x : 101
```

`System.out.println("x: " + x);` (Line 1)

- In the main method, x is initialized to 100.
- Output: x: 100

`increment(x);` (Line 2)

- The value of x (100) is passed into the increment method.
- Inside increment, `x++` increases the local variable x to 101.
- `System.out.println("x: " + x);` inside the method executes.
- Output: x: 101

`System.out.println("x: " + x);` (Line 3)

- In Java, primitive data types are passed by value. Changes made to x inside the increment method do not affect the original variable x in the main method.
- Therefore, x in main remains 100.
- Output: x: 100

`x = increment(x);` (Line 4)

- The `increment(x)` method is called again, printing x: 101 from inside the method. (Output: x: 101)
- This time, the returned value (101) is reassigned back to x in the main method (`x = ...`).

`System.out.println("x: " + x);` (Line 5)

- Now x in main has been updated to 101.
- Output: x: 101

Q27.

```
Q11.java ~ Q12.java ~ Q13.java ~ Calculator.java ~ Q15.java ~ Q16.java
1  import java.util.*;
2  class Q27 {
3      public static boolean countEvenDigit(int x){
4          while (x > 0) {
5              int digit = x % 10;
6              if (digit % 2 != 0) {
7                  return false;
8              }
9              x = x / 10;
10         }
11         return true;
12     }
13     public static void main(){
14         Scanner input=new Scanner(System.in);
15
16         System.out.print("Enter integer number : ");
17         int num = input.nextInt();
18
19         boolean alleven= countEvenDigit(num);
20         System.out.println(alleven);
21
22     }
23 }
24
25
```

Output>>

```
Enter integer number : 2343
false

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
ys
nt
oc
hi
Enter integer number : 28462
true

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q28.

Answers

B,C,D,G

Q29.

```
Q11.java × Q12.java × Q13.java × calculator.java × Q15.java × Q16
import java.util.*;
class Q29 {
    public static int getMonthDay(String x){
        switch (x){
            case "January", "january":
            case "March", "march":
            case "May", "may":
            case "July", "july":
            case "August", "august":
            case "October", "october":
            case "December", "december":
                return 31;
            case "April", "april":
            case "June", "june":
            case "September", "september":
            case "November", "november":
                return 30;
            case "February", "february":
                return 28;

            default:
                System.out.println("Month name incorrect !");
                return 0;
        }
    }

    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter Month Name : ");
        String month = input.nextLine();

        int days =getMonthDay(month);
        if(days !=0){
            System.out.println("Number of days : "+days);
        }
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd × +
Enter Month Name : january
Number of days : 31
```

```
C:\WINDOWS\SYSTEM32\cmd × +
Enter Month Name : sseptmber
Month name incorrect !
```

Q30.

```
import java.util.*;
class Q30 {
    public static void printOddAndEvenDigits(int number) {
        String oddDigits = "";
        String evenDigits = "";

        int reversed = 0;
        int temp = number;

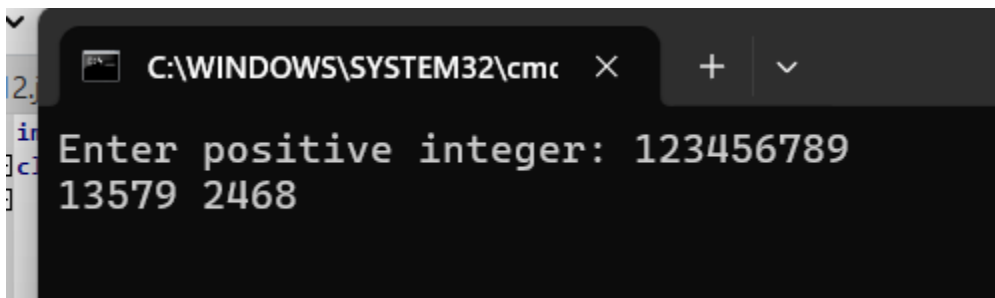
        if (temp == 0) {
            evenDigits = "0";
        }
        while (temp > 0) {
            int remainder = temp % 10;
            reversed = (reversed * 10) + remainder;
            temp /= 10;
        }
        while (reversed > 0) {
            int digit = reversed % 10;

            if (digit % 2 != 0) {
                oddDigits += digit;
            } else {
                evenDigits += digit;
            }
            reversed /= 10;
        }
        System.out.println(oddDigits + " " + evenDigits);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter positive integer: ");
        int num = input.nextInt();

        printOddAndEvenDigits(num);
    }
}
```

Output>>



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter positive integer: 123456789
13579 2468
```

Q31.

```
import java.util.*;
class Q31 {
    public static void printRolled(){
        Random r =new Random();
        int num1=r.nextInt(6)+1;
        int num2=r.nextInt(6)+1;
        int count = 0;
        while(num1!=num2){
            num1 = r.nextInt(6)+1;
            num2 = r.nextInt(6)+1;
            System.out.println(num1+" "+num2);
            count++;
        }
        System.out.println("Dice roll - "+count+" times");
    }
    public static void main(String[] args){
        printRolled();
    }
}
```

Output>>

```
6 2
4 6
6 1
6 3
4 1
5 1
5 3
2 5
5 2
6 6
Dice roll - 10 times
```

Q32.

Line 1 can inserted A,B,C,G

and line2 can inserted A,B,C,E,G

Q33.

Answers

C,D,E,F,H,K,O

Q34.

```
Example.java  Q27.java  Q29.java  Q30.java  Q31.java
import java.util.*;
class Q34 {
    public static long factorial(int num) {
        long fact = 1;
        for (int i = 1; i <= num; i++) {
            fact *= i;
        }
        return fact;
    }
    public static long calculateGroups(int n, int r) {
        if (r > n || n < 0 || r < 0) {
            return 0;
        }
        long nFactorial = factorial(n);
        long rFactorial = factorial(r);
        long nMinusRFactorial = factorial(n - r);

        return nFactorial / (nMinusRFactorial * rFactorial);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Number of children - ");
        int n = input.nextInt();

        System.out.print("Number of group members - ");
        int r = input.nextInt();

        long result = calculateGroups(n, r);
        System.out.println("The number of groups = " + result);
    }
}
```

Output>>

```
Number of children - 10
Number of group members - 3
The number of groups = 120
_____
```

Q35.

```
Q27.java x Q29.java x Q30.java x Q31.java x Q32.java x Q34.java x Q3
class Q35 {
    public static long findSmallestMultiple() {
        long number = 1;
        while (true) {
            boolean isDivisible = true;
            for (int i = 1; i <= 20; i++) {
                if (number % i != 0) {
                    isDivisible = false;
                    break;
                }
            }
            if (isDivisible) {
                return number;
            }
            number++;
        }
    }
    public static void main(String[] args) {
        long result = findSmallestMultiple();
        System.out.println("Smallest positive number: " + result);
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd x
Smallest positive number: 232792560

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q36.

```
import java.util.*;
class Q36 {
    public static void checkQuadrant(double x, double y) {
        if (x < -5 || x > 5 || y < -5 || y > 5) {
            System.out.println("Error: Coordinates must be between -5 and 5 as shown in the grid.");
            return;
        }
        if (x > 0 && y > 0) {
            System.out.println("Quadrant I");
        } else if (x < 0 && y > 0) {
            System.out.println("Quadrant II");
        } else if (x < 0 && y < 0) {
            System.out.println("Quadrant III");
        } else if (x > 0 && y < 0) {
            System.out.println("Quadrant IV");
        } else {
            System.out.println("Origin");
        }
        double sumOfSquares = (x * x) + (y * y);
        System.out.println("Distance: " + sumOfSquares);
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter x coordinate between (-5 to 5): ");
        double x = input.nextDouble();
        System.out.print("Enter y coordinate (-5 to 5): ");
        double y = input.nextDouble();

        checkQuadrant(x, y);
    }
}
```

Output>>

```
Enter x coordinate between (-5 to 5): 4
Enter y coordinate (-5 to 5): -5
Quadrant IV
Distance: 41.0
```

```
-----
(program exited with code: 0)
```

Q37.

```
import java.util.*;
class Q37 {
    public static void checkTriangle(double x, double y, double z) {
        if ((x + y > z) && (x + z > y) && (y + z > x)) {
            double perimeter = x + y + z;
            System.out.println("Valid triangle. Perimeter: " + perimeter);
        } else {
            System.out.println("Invalid triangle");
        }
    }

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter side x: ");
        double x = input.nextDouble();

        System.out.print("Enter side y: ");
        double y = input.nextDouble();

        System.out.print("Enter side z: ");
        double z = input.nextDouble();

        checkTriangle(x, y, z);
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter side x: 45
Enter side y: 54
Enter side z: 34
Valid triangle. Perimeter: 133.0
```

Q38.

```
import java.util.*;
class Q38 {
    public static void findCoordinates(double xA, double yA, double xB, double yB, double lambda, double mu) {
        double xC = (lambda * xB + mu * xA) / (lambda + mu);
        double yC = (lambda * yB + mu * yA) / (lambda + mu);

        System.out.println("Coordinates of C:");
        System.out.println("Xc: " + xC);
        System.out.println("Yc: " + yC);
    }

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter XA: ");
        double xA = input.nextDouble();

        System.out.print("Enter YA: ");
        double yA = input.nextDouble();

        System.out.print("Enter XB: ");
        double xB = input.nextDouble();

        System.out.print("Enter YB: ");
        double yB = input.nextDouble();

        System.out.print("Enter lambda (λ): ");
        double lambda = input.nextDouble();

        System.out.print("Enter mu (μ): ");
        double mu = input.nextDouble();

        findCoordinates(xA, yA, xB, yB, lambda, mu);
    }
}
```

Output>>

```
n Enter XA: 4
n Enter YA: 5
n Enter XB: 3
a Enter YB: 4
e Enter lambda (?): 4
n Enter mu (?): 4
n Coordinates of C:
a Xc: 3.5
n Yc: 4.5
a
n -----
a (program exited with code: 0)
```

Q39.

```
class Q39 {  
    public static void add(int a, int b) {  
        System.out.println("Sum of integers: " + (a + b));  
    }  
    public static void add(double a, double b) {  
        System.out.println("Sum of doubles: " + (a + b));  
    }  
    public static void add(String a, String b) {  
        System.out.println("Concatenated String: " + (a + b));  
    }  
    public static void main(String[] args) {  
        add(10, 20);  
        add(5.5, 4.2);  
        add("Hello, ", "World!");  
    }  
}
```

Output>>

```
Sum of integers: 30  
Sum of doubles: 9.7  
Concatenated String: Hello, World!  
  
-----  
(program exited with code: 0)  
  
Press any key to continue . . . |
```

Q40.

```
import java.util.*;
class Q40 {
    public static int absoluteDifference(int a, int b) {
        int diff = a - b;
        if (diff < 0) {
            return -diff;
        }
        return diff;
    }
    public static double absoluteDifference(double a, double b) {
        double diff = a - b;
        if (diff < 0) {
            return -diff;
        }
        return diff;
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = input.nextInt();
        System.out.print("Enter second integer: ");
        int num2 = input.nextInt();

        int intResult = absoluteDifference(num1, num2);
        System.out.println("Absolute difference of integers: " + intResult);

        System.out.print("Enter first double: ");
        double d1 = input.nextDouble();
        System.out.print("Enter second double: ");
        double d2 = input.nextDouble();

        double doubleResult = absoluteDifference(d1, d2);
        System.out.println("Absolute difference of doubles: " + doubleResult);
    }
}
```

Output>>

```
Enter first integer: 4
Enter second integer: -6
Absolute difference of integers: 10
Enter first double: 4.6
Enter second double: -0.7
Absolute difference of doubles: 5.3
```

Q41.

```
Q22.java Q23.java Q24.java Q25.java Example.java Q27.java
import java.util.*;
class Q41 {
    public static double getcal(double radius) {
        return 2 * 3.14 * radius;
    }
    public static double getcal(double length_s, double width_s) {
        if (length_s == width_s) {
            return 4 * length_s;
        } else {
            return (2 * (length_s + width_s));
        }
    }
    public static double getcal(double s01, double s02, double s03) {
        return s01 + s02 + s03;
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("--- circle ---");
        System.out.print("Enter radius of the circle : ");
        double c_radius = input.nextDouble();
        double c_cal = getcal(c_radius);
        System.out.println("Circle perimeter : " + c_cal);
        System.out.println();

        System.out.println("--- Square ---");
        System.out.print("Enter length : ");
        double sq_length = input.nextDouble();
        System.out.print("Enter width : ");
        double sq_width = input.nextDouble();
        double cal_valu = getcal(sq_length, sq_width);
        if (sq_length == sq_width) {
            System.out.println("Square perimeter : " + cal_valu);
        } else {
            System.out.println("Error: Length and width must be equal for a Square!");
        }
        System.out.println();
        System.out.println("--- Rectangle ---");
        System.out.print("Enter length : ");
        double rec_length = input.nextDouble();
        System.out.print("Enter width : ");
        double rec_width = input.nextDouble();

        if (rec_length != rec_width) {
            System.out.println("Rectangle perimeter : " + cal_valu);
        } else {
            System.out.println("Error: Length and width cannot be equal for a Rectangle!");
        }
        System.out.println();
        System.out.println("--- Triangle ---");
        System.out.print("Enter side 1 of the Triangle: ");
        double s1 = input.nextDouble();
        System.out.print("Enter side 2 of the Triangle: ");
        double s2 = input.nextDouble();
        System.out.print("Enter side 3 of the Triangle: ");
        double s3 = input.nextDouble();
        double tr_cal = getcal(s1, s2, s3);
        System.out.println("Triangle perimeter : " + tr_cal);
    }
}
```

Output>>

```
--- circle ---
Enter radius of the circle : 7
Circle perimeter : 43.96

--- Square ---
Enter length : 5
Enter width : 5
Square perimeter : 20.0

--- Rectangle ---
Enter length : 7
Enter width : 6
Rectangle perimeter : 20.0

--- Triangle ---
Enter side 1 of the Triangle: 5
Enter side 2 of the Triangle: 6
Enter side 3 of the Triangle: 6
Triangle perimeter : 17.0
```

Q42.

```
Q35.java  ✕  Q36.java  ✕  Q37.java  ✕  Q38.java  ✕  Q39.java  ✕  Q40.java  ✕  Q41.java
1  import java.util.*;
2  class Q42 {
3      public static boolean isPalindrome(int val) {
4          int original = val;
5          int flipped = 0;
6          while (val > 0) {
7              int rem = val % 10;
8              flipped = (flipped * 10) + rem;
9              val = val / 10;
10         }
11         return original == flipped;
12     }
13     public static boolean isPalindrome(String txt) {
14         String reversed = "";
15
16         for (int i = txt.length() - 1; i >= 0; i--) {
17             reversed = reversed + txt.charAt(i);
18         }
19
20         return txt.equals(reversed);
21     }
22     public static void main(String[] args) {
23         Scanner scan = new Scanner(System.in);
24
25         System.out.print("Enter an integer number: ");
26         int num = scan.nextInt();
27         System.out.println("Is number palindrome? : " + isPalindrome(num));
28
29         System.out.println();
30
31         System.out.print("Enter a word: ");
32         String str = scan.next();
33         System.out.println("Is word palindrome? : " + isPalindrome(str));
34     }
35 }
```

Output>>

```
Enter an integer number: 1221
Is number palindrome? : true

Enter a word: rur
Is word palindrome? : true

=====
```

```
Enter an integer number: 1232
Is number palindrome? : false

Enter a word: rush
Is word palindrome? : false

=====
```

Q43.

```
Q56.java Q57.java Q58.java Q59.java
import java.util.*;
class Q43 {
    public static String convertToBinary(int val) {
        String bin = "";
        while (val > 0) {
            int rem = val % 2;
            bin = rem + bin;
            val = val / 2;
        }
        if (bin.equals("")) {
            bin = "0";
        }
        return bin;
    }
    public static String convertToOctal(int val) {
        String oct = "";
        while (val > 0) {
            int rem = val % 8;
            oct = rem + oct;
            val = val / 8;
        }
        if (oct.equals("")) {
            oct = "0";
        }
        return oct;
    }
    public static String convertToHexadecimal(int val) {
        String hex = "";
        while (val > 0) {
            int rem = val % 16;

            if (rem == 10) {
                hex = "A" + hex;
            } else if (rem == 11) {
                hex = "B" + hex;
            } else if (rem == 12) {
                hex = "C" + hex;
            } else if (rem == 13) {
                hex = "D" + hex;
            } else if (rem == 14) {
                hex = "E" + hex;
            } else if (rem == 15) {
                hex = "F" + hex;
            } else {
                hex = rem + hex;
            }

            val = val / 16;
        }
        if (hex.equals("")) {
            hex = "0";
        }
        return hex;
    }
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        System.out.print("Enter a decimal number: ");
        int num = scan.nextInt();

        System.out.println("Binary: " + convertToBinary(num));
        System.out.println("Octal: " + convertToOctal(num));
        System.out.println("Hexadecimal: " + convertToHexadecimal(num));
    }
}
```

Output>>

```
Enter a decimal number: 200
Binary: 11001000
Octal: 310
Hexadecimal: C8
```

Q44.

```
Q37.java Q38.java Q39.java Q40.java Q41.java
import java.util.*;
class Q44 {
    public static int findFactorial(int val) {
        int fact = 1;

        for (int i = 1; i <= val; i++) {
            fact = fact * i;
        }
        return fact;
    }
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        System.out.print("Enter an integer number: ");
        int num = scan.nextInt();
        int re_fact = findFactorial(num);
        System.out.println("Factorial: " + re_fact);
    }
}
```

Output>>

```
Enter an integer number: 3
Factorial: 6
```

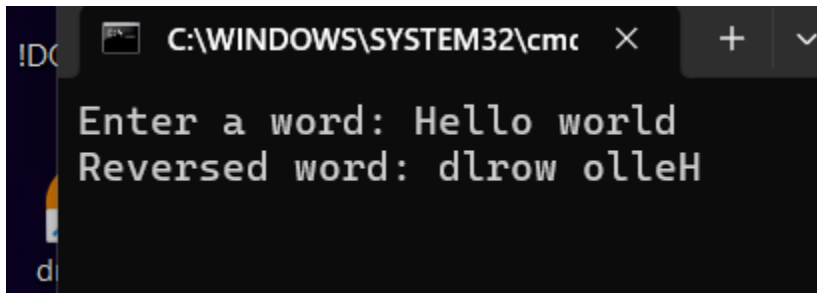
Q45.

```
import java.util.*;
class Q45 {
    public static String reverseString(String txt) {
        String reversed = "";

        for (int i = txt.length() - 1; i >= 0; i--) {
            reversed = reversed + txt.charAt(i);
        }
        return reversed;
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a word: ");
        String str = input.nextLine();
        String re_str = reverseString(str);
        System.out.println("Reversed word: " + re_str );
    }
}
```

Output>>



```
!DC C:\WINDOWS\SYSTEM32\cmd.exe
Enter a word: Hello world
Reversed word: dlrow olleH
```

Q46.

```
1  import java.util.*;
2  class Q46 {
3      public static void printGeometricSeries(long a, long r) {
4          long currentTerm = a;
5
6          for (int i = 1; i <= 10; i++) {
7              System.out.print(currentTerm + " ");
8              currentTerm = currentTerm * r;
9          }
10     }
11     public static void main(String[] args) {
12         Scanner input = new Scanner(System.in);
13
14         System.out.print("Enter 1st term (a): ");
15         long a = input.nextLong();
16
17         System.out.print("Enter common ratio (r): ");
18         long r = input.nextLong();
19
20         System.out.println("First 10 terms of the geometric series:");
21
22         printGeometricSeries(a, r);
23     }
24 }
```

Output>>

```
Enter 1st term (a): 6
Enter common ratio (r): 3
First 10 terms of the geometric series:
6 18 54 162 486 1458 4374 13122 39366 118098

-----
(program exited with code: 0)
```

Q47.

```
import java.util.*;
class Q47 {
    public static void power(int a,int b){
        int power_0 = (int)Math.pow(a,b);
        System.out.println("Result : "+power_0);
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        System.out.print("Enter base number      : ");
        int num1 = input.nextInt();

        System.out.print("Enter exponent (power) : ");
        int num2 = input.nextInt();

        power(num1,num2);
    }
}
```

Output>>

```
Enter base number      : 2
Enter exponent (power) : 3
Result : 8
```

Q48.

```
Q41.java  ✕  Q42.java  ✕  Q43.java  ✕  Q44.java  ✕  Q45.java  ✕  Q46.java
import java.util.*;
class Q48{
    public static int sumOfDigits(int num){
        int sum=0;
        while(num!=0){
            sum+=num%10;
            num/=10;
        }
        return sum;
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);
        System.out.print("Input an integer : ");
        int num=input.nextInt();

        int sum=sumOfDigits(num);
        System.out.println("Sum of digits for "+num+" : "+sum);
    }
}
```

Output>>

```
C:\WINDOWS\SYSTEM32\cmd  ✕  +  ▾
Input an integer : 2345
Sum of digits for 2345 : 14

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q49.

```
import java.util.*;
class Q49 {
    public static void tryguessing(int ra_num){
        Scanner input=new Scanner(System.in);

        for(int i=1;i<=5;i++){
            System.out.print("Enter your guessing number : ");
            int num0 = input.nextInt();
            if(num0==ra_num){
                System.out.println("Your Guessing is correct.");
            }else{
                if(num0 > ra_num){
                    System.out.println("Too high, try again.");
                }else{
                    System.out.println("Too low, try again.");
                }
            }
        }
        System.out.println("Maximum attempts exceeded ");
    }
    public static void main(String[] args){
        Scanner input=new Scanner(System.in);

        Random r = new Random();

        int r_num = r.nextInt(100)+1;

        tryguessing(r_num);
    }
}
```

Output>>

```
Enter your guessing number : 4
Too low, try again.
Enter your guessing number : 89
Too low, try again.
Enter your guessing number : 76
Too low, try again.
Enter your guessing number : 99
Too high, try again.
Enter your guessing number : 97
Too high, try again.
Maximum attempts exceeded
```

Q50.

```
Q42.java Q43.java Q44.java Q45.java Q46.java Q47.java
1 import java.util.*;
2 class Me {
3     public static void checkPassword() {
4         Scanner input = new Scanner(System.in);
5         String correctPassword = "1234";
6         int attempts = 0;
7
8         while (attempts < 3) {
9             System.out.print("Enter password: ");
10            String enteredPassword = input.nextLine();
11
12            if (enteredPassword.equals(correctPassword)) {
13                displayDetails();
14                return;
15            } else {
16                attempts++;
17                System.out.println("Incorrect password!");
18
19                if (attempts < 3) {
20                    System.out.print("Do you want to continue (Yes/No)? ");
21                    char choice = input.nextLine().charAt(0);
22
23                    if (choice=='N' || choice=='n') {
24                        System.out.println("Program ended.");
25                        return;
26                    }
27                }
28            }
29        }
30
31        System.out.println("Maximum attempts exceeded. Program ended.");
32    }
33
34    public static void displayDetails() {
35        System.out.println("\nLogin Successful!");
36        System.out.println("Name: Rush Wijemanna");
37        System.out.println("Age: 20");
38        System.out.println("Occupation: Software Engineer ");
39    }
40
41    public static void main(String[] args) {
42        checkPassword();
43    }
44 }
45
```

Output>>

```
Enter password: 13445
Incorrect password!
Do you want to continue (Yes/No)? y
Enter password: 12345
Incorrect password!
Do you want to continue (Yes/No)? n
Program ended.
```

```
Enter password: 245
Incorrect password!
Do you want to continue (Yes/No)? y
Enter password: 2345
Incorrect password!
Do you want to continue (Yes/No)? y
Enter password: 23456
Incorrect password!
Maximum attempts exceeded. Program ended.
```

```
Enter password: 1234

Login Successful!
Name: Rush Wijemanna
Age: 20
Occupation: Software Engineer
```